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# Habitat heterogeneity stimulates regeneration of bryophytes and vascular plants on disturbed minerotrophic peatlands

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Running head: Regeneration of disturbed minerotrophic peatlands

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**Abstract**

Wooded rich fens (WRF) are abundant in continental western Canada, characterized by high variation in surface topography and numerous plant species organized along microtopographic gradients. However, in regions where *in situ* oil sands exploration (OSE) prevails, winter operations eliminate the surface vegetation and mechanically flatten the exposed peat. This results in saturated or flooded soils during the growing season, and eliminates plant species dependent on naturally-elevated microhabitats with implications for peatland recovery. In northeastern Alberta, we re-developed hummock topography on replicate WRF after OSE by extracting blocks of frozen peat from peatland surfaces in winter. Peat mounds and adjacent unmounded flattened areas were left to regenerate naturally and were sampled four to five summers later. Mounds facilitated the colonization of many peatland plants not adapted to waterlogged soils. For bryophytes, mean richness and diversity of liverworts, *Sphagnum*, and true mosses were higher in mounded than unmounded plots. For vascular plants, woody plants (trees and shrubs) had higher richness, cover, and diversity (trees only) in mounded plots. Peat mounding may be effective for stimulating vegetation development on OSE-degraded WRF. All mounds, however, will require lateral expansion by hummock-forming mosses to provide the habitat volume required for development of large woody plants.

**Key-words**: microtopography, oil sands exploration, peatland restoration, *Sphagnum*, wooded rich fen

## Introduction

Peatlands are critically important wetland ecosystems because of the numerous functional and biological roles they have on landscapes (Vitt 2006; Minayeva et al. 2017). Peatlands account for approximately ⅓ of global soil carbon pools (Gorham 1991) and are particularly abundant in northern regions; more than one-third of global peatland area is found in Canada (Joosten and Clarke 2002). In continental western Canada, wooded rich fens (WRF) are one of the most abundant peatland types (Vitt et al. 1997). These fens are characterized in part by high variation in surface topography (Caners and Lieffers 2014), resulting in large numbers of bryophyte and vascular plant species that sort themselves along microtopographic gradients in relation to elevation above the mean water table position (Vitt et al. 1995).

Environmental heterogeneity is considered to be an important driver of species diversity in communities (MacArthur and Wilson 1967; Tilman and Pacala 1993). Higher numbers of habitats allow for coexistence of higher numbers of species through partitioning of niche space (habitat heterogeneity hypothesis; Ricklefs 1977; Palmer 1994). However, in regions where *in situ* oil sands exploration (OSE) is prevalent (Government of Alberta 2018), the natural hummock-hollow topography of peatlands, typical of WRF, is largely eliminated. This has resulted in the local elimination of many peatland plant species that depend on microhabitats found within the natural surface heterogeneity (Caners and Lieffers 2014).

OSE is a relatively recent disturbance type on the landscape, with the majority of drilling pads in the region having been developed since the mid-1990s (Lee and Boutin 2006). OSE operations create dense networks of temporary drilling pads (each approximately 0.7 ha) and ice roads used to explore deeply buried bitumen deposits. Exploration, including the construction of pads and roads, is carried out during winter when pads are frozen with concrete frost, needed to support the heavy machinery required for drilling. Winter exploration first eliminates all of the peatland surface vegetation and exposes the uppermost layers of non-vegetated peat, without reaching mineral soil. The surface is then mechanically compressed to develop a level surface and to promote deeper frost penetration prior to installation of a drill pipe located centrally on the pad. After drilling, which typically lasts a few days, the pads are cleaned up around the drill pipe and abandoned by spring. The flattened surfaces on abandoned pads are saturated and sometimes flooded with standing water during the growing season (Caners and Lieffers 2014). Standing water has been documented on drilling pads more than a decade after their abandonment (Caners and Lieffers 2014). The bare peat on abandoned drilling pads is rapidly colonized and dominated by sedge and bryophyte species that are tolerant of waterlogged soil conditions (Caners and Lieffers 2014), resembling closely the early stages of peatland development in continental western Canada (Kuhry et al. 1993). Macrofossil evidence has demonstrated that early peatlands can remain sedge-dominated for centuries to millennia (MacDonald 1987; Kuhry et al. 1993), suggesting that these OSE peatlands will be slow to recover to wooded fens. Increasing habitat heterogeneity by restoring hummocks on the flattened OSE drilling pads may be necessary for establishing the many specialized bryophyte and vascular plant species that are intolerant of saturated soils and require elevated positions above the usual water table. Increasing surface topography may also be applicable to other cold environments of the world where peatland surfaces are flattened and remain saturated, including removal of oil and gas infrastructures such as clay pads that were built on top of peatlands (Sobze et al. 2012; Shunina et al. 2016), or blocking of drainage channels in mined peatlands to re-establish hydrology and promote plant regeneration (Schimelpfenig et al. 2014; Malloy and Price 2014; Menberu et al. 2016). Despite the extensive areas affected by OSE, the continued growth of the sector, and the substantial alteration of vegetation on affected peatlands, there has been little literature on aided recovery of these degraded ecosystems (but see Lieffers et al. 2017 for tree species).

Until recently peatland restoration in North America has focused on ombrotrophic bogs after commercial peat extraction, whereas restoration of fens has proceeded more slowly, owing in part to greater complexity in restoring hydrology (Chimner et al. 2016; Rochefort et al. 2016). The majority of studies on fen restoration are related to the regeneration of vegetation following drainage for forestry or peat mining (reviews by Grand-Clement et al. 2015; Lamers et al. 2015; Haapalehto et al. 2017). In these systems, restoration usually involves blocking of drainage channels and reintroducing mosses and protecting them from desiccation (Rochefort et al. 2003). Peat mining removes the upper layers of peat in which viable propagules are concentrated, thus necessitating restoration via primary successional pathways (Campbell et al. 2003). By contrast, peatlands degraded by OSE can reestablish through secondary succession via persistent propagule banks in the surface peat layers or by aerial diaspore rain (Caners and Lieffers 2014). Reintroducing surface microtopography has been recognized as important for stimulating vegetation development in mined peatlands (Rochefort and Campeau 1997; Triisberg et al. 2014; citations in Chimner et al. 2016). However, restoration measures that are typically used for mined peatlands are problematic when applied to OSE-degraded peatlands with waterlogged or flooded soils. Successful measures to reintroduce the wide range of plant species typical of intact wooded fens following OSE remain poorly understood.

Within a large minerotrophic peatland complex in Alberta, Canada, we re-developed hummock topography on operational temporary OSE drilling pads before they were abandoned in winter. This was done by excavating and mounding the still-frozen peat on the flattened pad surfaces after drilling was complete. We examined the influence of mounds on the recovery of bryophytes and vascular plants as compared to immediately adjacent areas of flattened peatland, four to five summers after pads were abandoned. Based on observations from our previous study (Caners and Lieffers 2014), we hypothesized that 1) introduction of mounds on flattened drilling pads will provide habitat for the natural regeneration of native bryophyte and vascular plant species, especially those intolerant of saturated or flooded soil surface conditions, and 2) increased habitat heterogeneity provided by mounds on flattened drilling pads will increase the total number of WRF plant species that colonize drilling pads.

## Materials and methods

The study was conducted within a large minerotrophic peatland complex in northeastern Alberta, Canada, approximately 40 km southeast of the town of Conklin (**Fig 1**). A central drilling pad (2-32, number represents a legal land subdivision) was located at 55° 26’ 52.3” N, 110° 53’ 23.9” W. This peatland landscape is characterized by aeolian sand outcrops and variation in peatland type and water flow patterns (**Fig 1**). Study sites were classified as wooded moderate-rich fens based on surface water on drilling pads that ranged from pH 6.5 to 7.3 (mean = 6.9) and moderate alkalinity (mean = 94.7 mg $L^{-1}$ CaCO3; **Table 1**), and the presence of indicator bryophytes in the adjacent reference habitat that included minerotrophic *Sphagnum* species (e.g., *S. warnstorfii*, and *Sphagnum* species that are often abundant in wooded rich fens in the region, such as *S. angustifolium*), *Tomentypnum nitens* (Hedw.) Loeske, *Hamatocaulis vernicosus* (Mitt.) Hedenas, and *Brachythecium acutum* (Mitt.) Sull. (Chee and Vitt 1989; Vitt 1994; Alberta Environment and Sustainable Resource Development 2015). The adjacent reference habitat for study sites had a tree layer that was dominated by black spruce (*Picea mariana* (P. Mill) B.S.P.) or tamarack (*Larix laricina* (Du Roi) K. Koch). Climate of the region is continental with mean daily temperature 2.1°C and mean total annual precipitation 421 mm (Environment and Climate Change Canada 2018; Cold Lake A meteorological station).

Field sampling

We selected five temporary drilling pads (each approximately 0.7 ha; furthest distance between pads is 4.3 km) that were constructed by Devon Energy Corporation during winter 2011–2012 as part of their standard *in situ* oil sands exploration (OSE) operations to assess the depth and quality of deeply buried bitumen reserves. On these drilling pads, trees were cut and either mulched (dispersed) or windrowed to the sides of pads. The snow and remaining vegetation (including tree stumps and roots, and living bryophyte layer) were repositioned by blading, exposing the uppermost layers of non-vegetated peat (**Fig 2**) without reaching mineral soil. The exposed peat was then compacted by machine traffic and some pads had water added to allow full penetration of concrete frost into the peat soils. The level pads and ice roads leading to them were consequently strong enough to support the heavy machinery and drilling rigs required for drilling.



Shortly after exploratory drilling all machinery were removed and the frozen pad surface was sliced into a grid (1.5 m x 1.5 m squares) using a single tooth shank ripper mounted on a bulldozer (D6 Caterpillar). A tracked excavator (Komatsu PC200LC) fitted with a standard bucket and thumb lifted out the frozen squares (35–60 cm thick, **Fig 2A, B**) and placed these next to their excavated holes with the upper peat layer positioned upright (**Fig 2C**). The size of mounds was kept as consistent as possible by machine operators. There was some variation in mound formation among the drilling pads: two pads (6-5, mounded on calendar day 35; 8-28 on day 76) had mounds that were removed as intact blocks of frozen peat; the other three pads (7-32, mounded on calendar day 67; 2-32 on day 70; and 3-33 on day 76) had peat that unexpectedly fractured on extraction (**Fig 2D**). All mounds were left to regenerate naturally from wind-dispersed seeds and spores from nearby intact habitat, or from bryophyte or root fragments and plant diaspores occurring naturally within the peat mounds and peatland surface.

In the fourth summer after mound preparation, mean mound height was 31.6 ± 4.3 (SE) cm and mean mound area was 5.2 ± 0.7 (SE) $m^2$ among drilling pads (**Table 1**). Von Post peat humification of mounds ranged from H = 2 (pad 8-28) to H = 5 (pad 2-32). Mounds were generally rounded in shape; however, mounds formed from intact blocks of frozen peat were slightly more angular in structure, with more vertical sides and more horizontal apices than mounds formed from fractured peat. Pads with the lowest mounds (i.e., 7-32 and 2-32) also had the lowest values for water pH, EC, and alkalinity, and appeared to be wetter, with shorter trees and higher densities of tamarack than black spruce.

In either the fourth or fifth growing season after mounding, we randomly sampled five mounds per pad (**Fig 2E, F**). Drilling pads were remote with difficult access, preventing the sampling of all sites in the same season. The outline of each mound was measured to estimate its area and was subsequently described as a mounded plot. Immediately adjacent to each mound the equivalent area of flattened, unmounded habitat was sampled similarly for plants, and was subsequently described as an unmounded plot. The water-filled excavated hollows were not sampled as they contained relatively few species at low abundance; we observed that these species were usually also present within unmounded areas. The water table position at the time of sampling was within 5 cm of the peatland surface on all drilling pads as determined by water levels in the excavated hollows adjacent to mounds. Within both mounded and unmounded plots, the abundance of each bryophyte and vascular plant species was estimated using vertically projected ocular cover. Plants that could not be identified confidently in the field were collected and identified in the lab. Collected specimens are held at the Royal Alberta Museum herbarium. Taxonomy is based on VASCAN (Brouillet et al. 2010+) for vascular plants, Flora of North America Editorial Committee (2007, 2014) for mosses, and Stotler and Crandall-Stotler (2017) for liverworts.

Water samples were collected from natural pools near the centre of pads in 1L Nalgene bottles. Samples were frozen prior to analysis by the Natural Resources Analytical Laboratory (NRAL) at the University of Alberta, for pH, electrical conductivity ($\mu s\ cm^{-1}$), and alkalinity ($mg\ L^{-1}$ $CaCO_3$). Analytical techniques used are available at https://nral.ualberta.ca/Analytical-Methods/. Methods to assess the degree of peat humification (H, using the von Post classification on a scale of H1 [completely undecomposed peat] to H10 [completely decomposed peat]) within mounds on each drilling pad, and the average height and density of conifer trees adjacent to each drilling pad are described in Lieffers et al. (2017). The height of each mound above the average adjacent substrate was measured using a horizontal bar and vertical rule, set perpendicular to each other with a hand level.



Statistical analyses

Differences in mean species richness, percent cover, and diversity between mounded and unmounded plots were tested separately for bryophytes and vascular plant species groups at the pad-level (n = 5) using paired t-tests (mounded and unmounded plots within a drilling pad were treated as sub-samples) in SPSS 24 (IBM SPSS Statistics). Plant species groups were defined on the basis of growth form. For bryophytes, growth form groups were liverworts, *Sphagnum*, and true mosses (i.e., those belonging to Class Bryopsida, but also including a few species belonging to Class Polytrichopsida to facilitate data interpretation; **Appendix Table A1**). For vascular plants, growth form groups were forbs, graminoids, shrubs, and trees.

Species richness was measured as the total number of species, and cover as the total ocular cover of species, on a mounded plot or adjacent unmounded plot. Diversity was calculated as the exponential of Shannon entropy (N1; Hill 1973), exp(H), where $H = -\Sigma p_i(\ln p_i)$ and $p_i$ is the proportional abundance of each species in a sample, estimated using percent cover, and the summation term is from i to S, where S is the total number of species per plot. The exponential of Shannon entropy provides many desirable properties as a diversity measure, and can be interpreted as the number of species in a sample had all species been equally common. The measure attains a maximum value equal to the total number of species in the sample. Each variable was Box-Cox transformed as needed to meet normality prior to analysis.

Species composition was compared between mounded and unmounded plots separately for bryophytes and vascular plants at the pad-level ($n = 5$) using Principal Coordinate Analysis (PCoA). Percent cover data for each species was log-transformed as $x' = \ln(x+1)$ to downweight the influence of dominant species and was represented using Bray-Curtis dissimilarity. The hilltop plot technique (Nelson et al. 2015) was used to display multiple non-linear response surfaces within the PCoA ordination space. Each "hilltop" is based on a response surface of total percent cover values for each bryophyte or vascular plant growth form group. The response surface is calculated by interpolating the cover values for a group among the $n = 10$ drilling pads in the ordination space. For each group, the hilltop represents the top 20% of the interpolated range of values; i.e., the highest total cover values for that group of species (Nelson et al. 2015). Differences in species composition between mounded and unmounded plots were tested by Permutational Multivariate Analysis of Variance (PERMANOVA) at the pad-level using the same data as PCoA, with 9999 permutations for tests of significance. Indicator Species Analysis

(Dufrêne and Legendre 1997) identified species that were closely affiliated with either mounded or unmounded habitat. Analyses were performed in PC-ORD 7.02 (McCune and Mefford 2016).

Mound height was expected to influence the numbers of species on mounds. Richness of species belonging to each plant growth form group was compared to mound height by Spearman rank-correlation in SPSS 24 (IBM SPSS Statistics).

## Results

A large number of bryophyte and vascular plant species was observed on mounded and unmounded plots on drilling pads. These species are characteristic of intact WRF of the region, where they grow at different microtopographic positions in relation to the mean water table position (Caners and Lieffers 2014). A total of 66 bryophyte species occurred on mounds and 44 species on unmounded areas. For vascular plants, there was a total of 70 species on mounds and 62 species on unmounded areas (**Appendix Table A1**). Venn diagrams highlight the important contribution of mounds to total species numbers across the study system for bryophytes and vascular plants (**Fig 3**). For bryophytes, there was a total of 26 species found on mounded plots only, 40 on both mounded and unmounded plots, and 4 on unmounded plots only. For vascular plants, there was a total 16 species found on mounded plots only, 54 on both mounded and unmounded plots, and 8 on unmounded plots only.

Mounding had a substantial influence on the richness, cover, and diversity of bryophytes and vascular plants across the five OSE pads (**Fig 4**). For bryophytes, mean richness and diversity of liverworts, *Sphagnum*, and true mosses were significantly higher on mounded plots than unmounded ones. Mean cover of liverworts was higher on mounded plots, whereas cover of true mosses was higher on unmounded plots. For vascular plants, mean richness and cover of shrubs and trees was higher on mounds, but mean richness of graminoids was higher for unmounded plots. Mean diversity of trees was higher on mounded than unmounded plots; the other plant growth form groups did not demonstrate a response.

Composition of bryophytes differed significantly between mounded and unmounded plots (**Fig 5**, **Appendix Table A2**). The first two PCoA axes combined explained a total of 69.4% of variation in species composition, with mounded and unmounded groups being separated along PCoA axis 1. Some species in the ordination space had centroids that were associated with unmounded areas. These are characteristic rich fen bryophytes that are aquatic or adapted to saturated soils and included (in order of increasing centroid values on PCoA axis 1) *Scorpidium revolvens*, *Drepanocladus aduncus*, *Calliergon richardsonii*, *Calliergon giganteum*, *Hamatocaulis vernicosus*, *Paludella squarrosa*, *Drepanocladus sordidus*, *Tomentypnum nitens*, and *Meesia longiseta*. A larger number of bryophyte species had centroids that were associated with mounds. Hilltop overlays in the PCoA ordination space for total liverwort and *Sphagnum* cover were associated with mounded plots only for drilling pads, but the hilltop overlay for true mosses spanned both the mounded and unmounded plots. Most (14 of 15) bryophytes detected by Indicator Species Analysis (ISA; **Appendix Table A3**) were associated with mounded plots and included several taxa that occur frequently in natural wooded fens in the region (Caners and Lieffers 2014), such as *Polytrichum strictum*, *Dicranum undulatum*, *Pohlia nutans*, *Aneura pinguis*, *Fuscocephaloziopsis pleniceps*, *Polytrichum juniperinum*, *Cephaloziella rubella*, and *Fuscocephaloziopsis lunulifolia*. The only taxon from the ISA associated with unmounded plots was *Hamatocaulis vernicosus*.

Composition of vascular plants did not have as pronounced a response between mounded and unmounded plots as compared to bryophytes (**Fig 6, Appendix Table A2**). The first two PCoA axes combined explained a total of 58.2% of variation in species composition, with mounded and unmounded groups separated along PCoA axis 2. Several species had centroids that were associated with unmounded plots for drilling pads. These species are mostly aquatic or adapted to saturated soils and included (in order of decreasing centroid values on PCoA axis 2) *Carex canescens*, *Triantha glutinosa*, *Utricularia intermedia*, *Sarracenia purpurea*, *Agrostis scabra*, *Ranunculus gmelinii*, *Eriophorum viridi-carinatum*, *Typha latifolia*, *Triglochin maritima*, and *Epilobium palustre*. There were several species with centroids associated with mounded plots, including many shrubs and trees. Hilltop overlays in the PCoA ordination space for total tree and shrub cover were associated with mounds, whereas hilltop overlays for graminoids and forbs were associated with unmounded plots. The ISA for vascular plants detected relatively few indicators for mounded or unmounded plots (**Appendix Table A3**). All of the taxa detected for mounded plots were woody plants. A single taxon was associated with unmounded plots, the aquatic *Triglochin maritima*.

Mound height was positively correlated with true moss richness ($\rho = 0.49$, $p = 0.013$) and was negatively correlated with graminoid richness ($\rho = -0.43$, $p = 0.031$) and shrub richness ($\rho = -0.43$, $p = 0.033$) (**Table 2**). Shrub diversity was negatively correlated with shrub cover ($\rho = -0.53$, $p = 0.007$) and graminoid cover was negatively correlated with true moss cover ($\rho = -0.60$, $p = 0.001$).

## Discussion

Mounding increased peatland habitat heterogeneity

As predicted by habitat heterogeneity theory (e.g., Vivian-Smith 1997), the increased surface topography and diversity of habitats provided by mounds facilitated the colonization of a wider range of peatland plants than unmounded habitat alone on these *in situ* oil sands exploration (OSE) pads. Elevated mounded substrate on drilling pads acted as "habitat islands" for many vascular plant and bryophyte species that are not adapted to very wet growing conditions. Mounds with their elevated habitats could provide a "bet-hedging" strategy (sensu Pastorok et al. 1997; Doherty and Zedler 2015), by elevating some species above high water levels that can occur after spring snow melt or heavy rainfall during summer. Mounding may, therefore, be an effective method to stimulate vegetation establishment on drilling pads if the mounds can persist and remain elevated above the water table. However, both mounded and unmounded habitat will be required for the regeneration of the numerous plant species documented on drilling pads.

Mounding influenced plant species richness, cover, diversity, and composition

Mounds were important for supporting a higher richness, cover, and diversity of several plant groups as compared to adjacent flattened areas. More of the species detected were found exclusively on mounds as compared to the few species found exclusively on the saturated unmounded soils. For species found in both mounded and unmounded plots, many had lower frequency in unmounded plots. Unmounded habitat was characterized by fewer species and lower diversity; the true moss *Hamatocaulis vernicosus* was particularly frequent and abundant because of its affinity for wet rich fen soils (Borkenhagen and Cooper 2018). The combination of mounded and unmounded habitat on OSE pads appears to mimic partly the hummock-hollow topography of natural WRF, providing a wider range of habitat conditions for plant establishment. Most of the bryophyte diversity on pads was found on the lower edges of mounds similar to the transitions from hollow to hummock positions noted in similar fen types (Gignac 1992).

The composition of bryophytes differed significantly between mounded and unmounded areas, whereas differences in composition of vascular plants were less pronounced based on PERMANOVA. The larger difference for bryophytes as compared to vascular plants is consistent with the high affinity of many bryophyte species for fine-scale microhabitats in peatlands (Gignac 1992; Vitt et al. 1995). Many bryophytes are unable to establish or persist under saturated or flooded conditions (Caners and Lieffers 2014) making mounds potentially important for their representation. Borkenhagen and Cooper (2018) examined the effects of submergence on four fen mosses and found *Hamatocaulis vernicosus* was the only species that did not decline in the short-term (after 6 weeks) or long-term (after 11 months). *Tomentypnum nitens* and *Aulacomnium palustre* exhibited short-term declines but recovered partially in the long-term, whereas *Sphagnum warnstorfii* was particularly susceptible and decreased without recovery after two weeks.

Differences in bryophyte and vascular plant species composition among the drilling pads were observed along the axes of the indirect gradient analysis and may be partly attributable to mound height. The highest mounds in the fourth summer after pad preparation (i.e., those on pads 8-28 and 6-5) were extracted in winter as intact blocks of frozen peat, whereas pads with smaller mounds (pads 3-33, 7-32, and 2-32) were made of peat that shattered into smaller fragments on extraction. Mounds made of fragmented peat may have been more susceptible to erosion by heavy summer rains, spring snowmelt, and winter frost heaving. Further, mounds composed of highly humified peat may have been more susceptible to erosion than mounds composed of peat with lower humification (Tuukkanen et al. 2014). Although pads 8-28 and 6-5 had similar mounds made of intact peat, they were widely separated in the ordination space in terms of bryophyte composition on mounds. This separation likely resulted from higher mean cover of bryophytes on mounds at pad 8-28 and several species on mounds that differed between the pads; these pads were the most widely separated geographically.

The moisture content within mounds may be related to mound height and may influence the species that can establish on them. The tops of the tallest mounds were drier than the lower edge portions of mounds that were mesic and closer to the pad surface (R.T. Caners, personal observation). The higher richness of true mosses with increasing mound height may reflect a stronger moisture gradient on taller mounds, and correspondingly higher number of microhabitats to support a wider range of moss species. The negative relationship between graminoid cover and true moss richness suggests that high graminoid abundance can limit the presence of these bryophytes. In comparison, shrub richness decreased with increasing mound height; this may be attributed to drier conditions at the tops of the tallest mounds, or because of higher dominance of a few shrub species.

Unlike bryophytes, vascular plant composition for both mounded and unmounded habitats exhibited high variation in species composition among drilling pads along the first ordination axis. Drilling pads differed in terms of their regional position within the landscape and differed in overall wetness (Lieffers et al. 2017). Vascular plants likely colonized the mounds via vegetative expansion of rhizomatous species within the peat, and establishment of seed from seedbanks and by dispersal from treed areas nearby. The combination of site-level differences among drilling pads together with the more limited dispersal capacities of vascular plant seeds and rhizomes as compared to bryophyte spores and asexual propagules, likely contributed to the observed differences in vascular plant composition among drilling pads.

Although natural differences in vegetation composition among drilling pads appeared to be a driver of vascular vegetation establishment in mounded and unmounded areas, the mounding treatments also had discernable effects; these are best characterized by patterns in the abundance and dominance of woody plants and *Carex* species (sedges). Vascular plants in the unmounded plots were heavily dominated by *Carex* species (e.g., *C. aquatilis*, *C. diandra*, *C. disperma*, *C. gynocrates*, *C. heleonastes*, *C. limosa*, *C. magellanica*, C. *prairea*, *C. rostrata*, *C. tenuiflora*), which composed nearly 80% of the vascular plant cover within these areas. By contrast, mounds were characterized by a higher abundance of woody plants, primarily *Salix* species (willows), *Betula* species (birches), and ericaceous shrubs. Woody species were co-dominant with *Carex* species on the mounds, but were nearly absent in the unmounded plots. These patterns are likely attributable to the strong differences in soil moisture between mounded and unmounded areas, with *Carex* species dominating in waterlogged soils in unmounded areas and woody plants dominating in the drier habitat of the mounds. A similar pattern is commonly observed in wetlands throughout the boreal biome, where *Carex* species are typically most abundant on saturated soils or in shallow standing water, while shrubs are normally more prevalent in drier areas near the wetland margin (Cronk and Fennessy 2001). *Carex* are generally not reliant on mycorrhizae, which do not tolerate persistent anaerobic conditions (Rydin and Jeglum 2006). By contrast, many tree and shrub species in boreal peatlands are not adapted to permanently wet soils (Shaw et al. 1990; Cronk and Fennessy 2001; Rydin and Jeglum 2006). This restricts woody plants to habitats above the typical water table, such as hummock tops (Lieffers et al. 2017).



The sources of plant propagules that colonized the mounded and unmounded plots on drilling pads is largely unknown. Propagules may have arrived as wind-dispersed seeds and spores from nearby intact habitat, or as bryophyte or root fragments and plant diaspores that occurred naturally within the peat. OSE drilling pads are relatively small in size (each approximately 0.7 ha) and almost completely surrounded by intact habitat, limiting the distance that seeds and spores arriving by wind would need to disperse. Tamarack and black spruce seedlings documented on drilling pads would have arrived as wind-dispersed seed from nearby habitat as the winged seeds of conifers are expected to fall within close proximity to parent plants (Lieffers et al. 2017). Similarly, trembling aspen (*Populus tremuloides* Michx.) and balsam poplar (*Populus balsamifera* L.) seedlings on mounds would have arrived by seed but from more distant sources, as these species do not occur in the peatland complex used for this study. Propagule banks within the exposed peat may also be an important mechanism for plant establishment on drilling pads. *Sphagnum* can form a long-term spore bank in bogs (Sundberg and Rydin 2000) and can regenerate from specialized (totipotent) cells in gametophytic tissue (Melosik and Såstad 2005), and bryophyte and vascular plant species characteristic of wooded fens have been shown to germinate from peat cores obtained from minerotrophic pealands (Miller et al. 2015). Studies on the sources of plants that establish on OSE pads would be insightful.

Differences in establishment mechanisms between woody plants and *Carex* might have also influenced plant composition on and off the mounds. Many of the sedge species found on the drilling pads, including the dominant *Carex aquatilis*, are rhizomatous, and likely colonized the pads after disturbance by vegetative expansion. Woody plants on mounds, by comparison, established either by seed that arrived by wind, or growth of root systems that remained viable within the peat after OSE. All tree and shrub species documented on mounds produce relatively small seeds with limited reserves and small germinants. Establishment and survival of woody plants was likely impeded on the tallest mounds by sedge litter and drier conditions. In nonmounded areas, limiting factors likely included litter buildup (Groot and Adams 1994; Greene et al. 1999), flooding (Kozlowski 1997), and competition with established *Carex* (Cronk and Fennessy 2001).

On mounds, the root networks of trees and shrubs should stabilise the mounds and reduce further erosion (Laberge et al. 2013; Pouliot et al. 2011*a*, 2011*b*), thereby maintaining the drier, aerated soil conditions that these plants require. However, continued growth and perhaps the long-term survival of trees on mounds will ultimately require expansion of root systems beyond the mound volume. The addition of elevated habitat for roots will necessitate lateral expansion of mounds by growth of hummock-forming *Sphagnum* (e.g., *Sphagnum warnstorfii*, *Sphagnum fuscum*) that grow at the base of mounds. These species form hummocks in natural intact wooded fens, and provide substrates for tree roots (Caners and Lieffers 2014). The fact that these *Sphagnum* species occurred frequently on mounds suggests they could expand the mounds, but this remains to be fully investigated. The moss *Polytrichum strictum* is strongly affiliated with mounds and may facilitate *Sphagnum* establishment on mounds through its influence as a nurse species (Groeneveld et al. 2007).

**Implications for peatland restoration**

The flattened and saturated areas of bare peat left behind after the preparation of temporary drilling platforms on frozen peatlands can be partly amended by developing mounded surfaces. Mounding may also be applicable to similar disturbances following the preparation of ice roads and seismic lines in winter, or in other environments where peatland surfaces are compressed and remain saturated during the growing season. By producing a range of habitats in relation to the usual water table position, mounding promotes the regeneration and natural sorting of wooded fen species along microtopographic gradients.

Mounds are more likely to experience less erosion and remain elevated over time if they are constructed from large blocks of intact frozen peat, especially if they are composed of peat with low levels of humification. Extracting these large blocks intact will require prolonged cold winter temperatures to allow for deeper frost penetration, and possibly lower peat humification to provide more structural integrity during and after extraction. Mounds composed of shattered peat fragments will erode more rapidly and may not provide the habitat volume required for root development and growth of woody plants. For all mound types, however, lateral expansion of mounds by hummock-forming mosses will be required eventually to expand the volume of rooting habitat for the woody plants growing on them.



Mounds provided elevated habitat needed by many bryophyte and vascular plant species (especially woody plants) that are intolerant of saturated soils and flood-averse, similar to patterns found in natural wooded fens of the region (Caners and Lieffers 2014). Creating mounds on OSE pads may be cost effective considering: 1) machinery used to produce mounds are generally available when pads are being decommissioned in winter, and 2) regeneration of the wide range of bryophytes and vascular plants on mounds occurs spontaneously and may not require seeding or planting of remote sites with difficult access.

In unmounded areas, *Carex* is expected to remain dominant for the foreseeable future. The wetter conditions without elevated substrates will continue to support species adapted to waterlogged soils, until *Carex* peat accumulates sufficiently to elevate the ground surface above the water level. Macrofossil studies in continental western North America peatlands have shown it can take centuries to switch from sedge- to *Sphagnum*-dominated peatlands during the developmental sequence from rich fen to poor fen to ombrotrophic bog (Kuhry et al. 1993).

Furthermore, mounding may contribute meaningfully to habitat recovery of the nationally threatened woodland caribou in the region (Environment Canada 2012; Pigeon et al. 2016). Establishment of trees and shrubs on mounds, coupled with the establishment of the mosses that continue to add to their substrate, has the potential to produce the visual screening that is an important attribute of caribou habitat.

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## Appendix A.

See Tables A1–A3.

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**Table 1**. Mound, reference habitat, and water chemistry characteristics for drilling pads examined in this study (modified from Lieffers et al. 2017). Mean mound height (SE [cm]) and area (SE [$m^2$]) were measured four summers after pad preparation. Mound peat form indicates whether peat was intact or fractured on extraction for mound construction. Date is the annual date of pad completion in 2012. Von Post peat humification (H) of mounds is the median of four mounds measured per pad. Spruce and tamarack density (trees $ha^{-1}$) and height (m) of co-dominants in adjacent intact habitat. Water pH, electrical conductivity (EC; µs $cm^{-1}$), and alkalinity (mg $L^{-1}$ $CaCO_3$); EC was adjusted for the effects of hydrogen ions.

| Pad | Mound ht. cm | Mound area $m^2$ | Mound peat form | Date | H | Spruce density trees $ha^{-1}$ (ht [m]) | Tamarack density trees $ha^{-1}$ (ht [m]) | pH | EC µs $cm^{-1}$ | Alkalinity mg $L^{-1}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 8-28 | 40.8 (2.1) | 3.8 (0.5) | intact | 76 | 2 | 2000 (8) | 2000 (8) | 6.9 | 203 | 99.6 |
| 6-5 | 39.8 (2.0) | 7.4 (1.8) | intact | 35 | 4 | 2000 (10) | 2000 (9) | 7.3 | 250 | 147.1 |
| 3-33 | 34.6 (5.2) | 5.8 (0.7) | fractured | 76 | 4 | 1500 (7) | 1500 (8) | 7.1 | 243 | 137.1 |
| 7-32 | 22.4 (2.7) | 3.3 (0.6) | fractured | 67 | 3 | 200 (8) | 1500 (6) | 6.8 | 84 | 45.2 |
| 2-32 | 20.6 (2.3) | 5.7 (0.3) | fractured | 70 | 5 | 400 (5) | 1500 (6) | 6.5 | 74 | 44.5 |

**Table 2**. Spearman rank-correlations (ρ) between mound height and plant richness, calculated separately for the different plant growth form groups for bryophytes and vascular plants. Calculations are based on individual mounds (n = 25).

| | **Mound height** (cm) | |
|---|---|---|
| | ρ | p-value |
| **Bryophyte richness** | | |
| Liverworts | 0.14 | 0.491 |
| True mosses | 0.49 | 0.013 |
| *Sphagnum* | 0.11 | 0.602 |
| Total | 0.40 | 0.048 |
| **Vascular plant richness** | | |
| Forbs | -0.17 | 0.422 |
| Graminoids | -0.43 | 0.031 |
| Shrubs | -0.43 | 0.033 |
| Trees | 0.39 | 0.053 |
| Total | -0.36 | 0.076 |

## Figure captions

**Figure 1**. Location of the five operational drilling pads used for this study in northeastern Alberta. Image dated 15 May 2015, from Google Earth Pro 7.3.

**Figure 2**. Examples of drilling pad features. A) Winter construction of pad 6-5, February 2012. B) Replicate drilling pads, June 2012. Water-filled hollows from which peat was extracted are visible in a checkerboard pattern. C) Mounds of intact peat on pad 8-28, June 2012. D) Mounds of fractured peat on pad 3-33, June 2012. E) Mounds of intact peat on pad 8-28, June 2016. F) Mounds of fractured peat on pad 2-32, June 2016.

**Figure 3**. Total numbers of vascular plant (left) and bryophyte (right) species across the study system by plant growth form group. Each diagram depicts the total number of species documented within mounds only (unshared area within thicker circles), unmounded habitat only (unshared area within thinner circles), and both mounded and unmounded habitat (shared area between thicker and thinner circles).



**Figure 4**. Mean (±1SE) species richness, percent cover, and diversity (N1) for bryophyte (top row) and vascular plant (bottom row) growth form groups. Results are based on n = 5 drilling pads. Shaded bars refer to mounded (M) plots; unshaded bars refer to unmounded (U) plots. Significant differences between mounded and unmounded plots for a plant group are indicated by asterisks: * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$. Abbreviations: gramin. = graminoids.

**Figure 5**. Results of indirect (PCoA) ordination of 70 bryophyte species among drilling pads for mounded and unmounded plots. Mounded and unmounded groups are labelled and bounded by polygons; group centroids are indicated as “+” symbols. Species codes are based on Appendix Table A1. The first two PCoA axes explained a total of 69.4% of variation in species

composition. Hilltop overlays are colour coded for the three bryophyte growth form groups (liverworts, *Sphagnum*, and true mosses) and represent the region in the ordination space with the highest (top 20%) total cover values for a group.

**Figure 6**. Results of indirect (PCoA) ordination of 78 vascular plant species among drilling pads for mounded and unmounded plots. Mounded and unmounded groups are labelled and bounded by polygons; group centroids are indicated as "+" symbols. Species codes are based on Appendix Table A1. The first two PCoA axes explained a total of 58.2% of variation in species composition. Hilltop overlays are colour coded for the four vascular plant growth form groups (forbs, graminoids, shrubs, and trees) and represent the region in the ordination space with the highest (top 20%) total cover values for a group.

Figure 1.

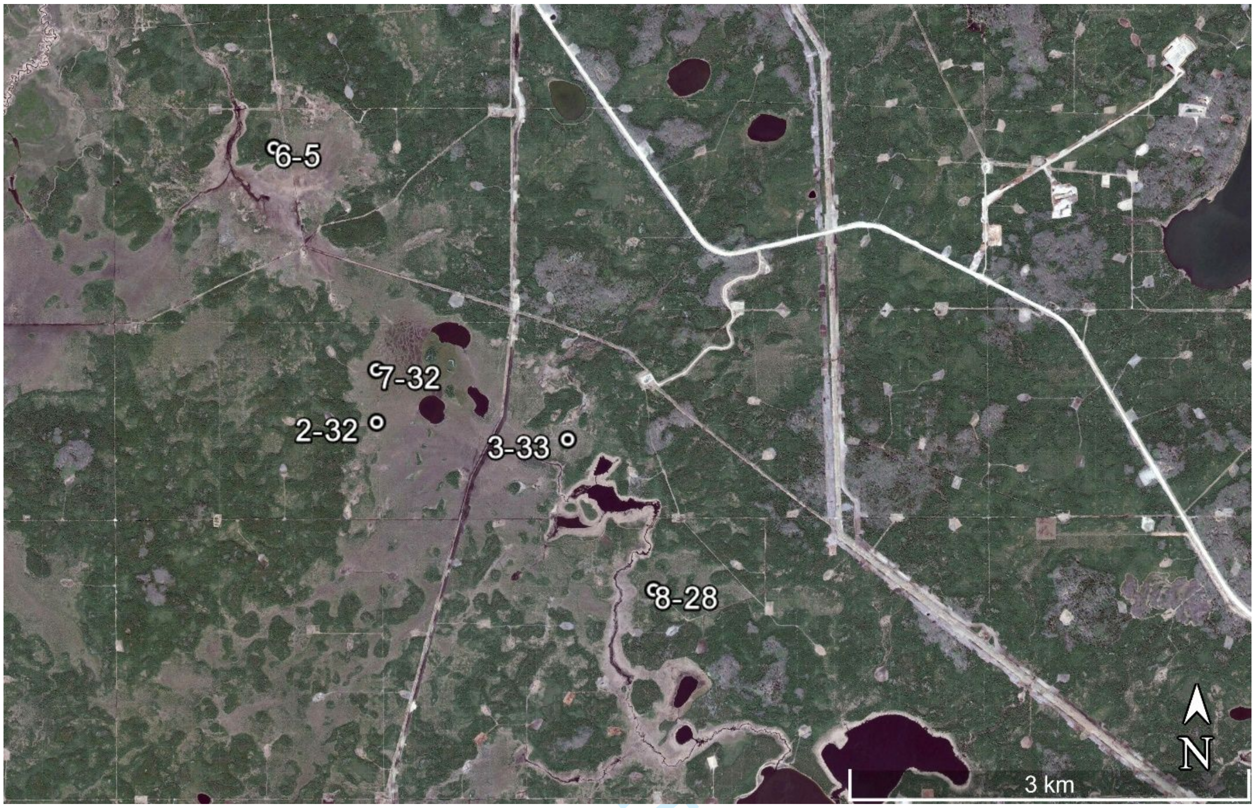


Figure 2.

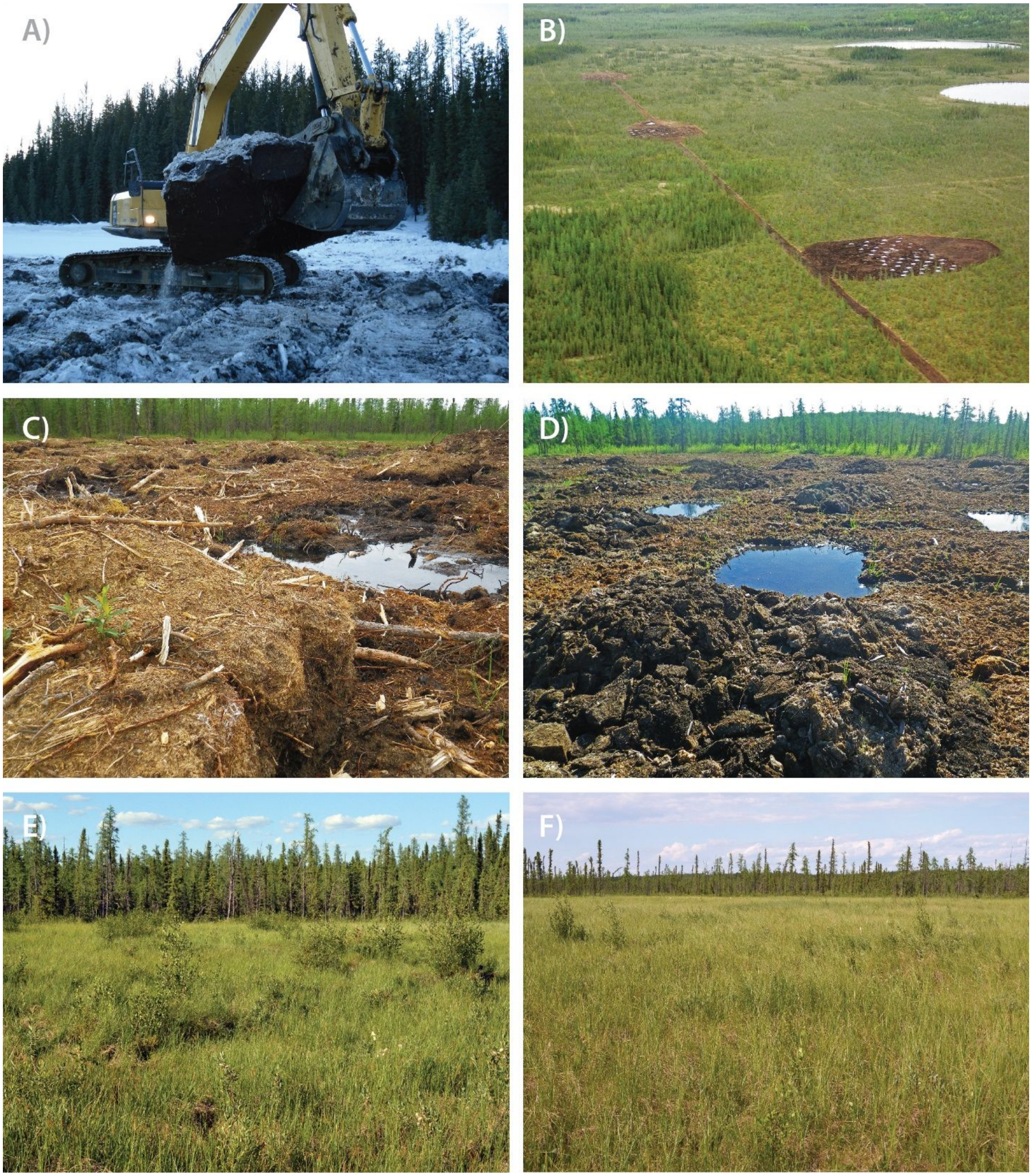
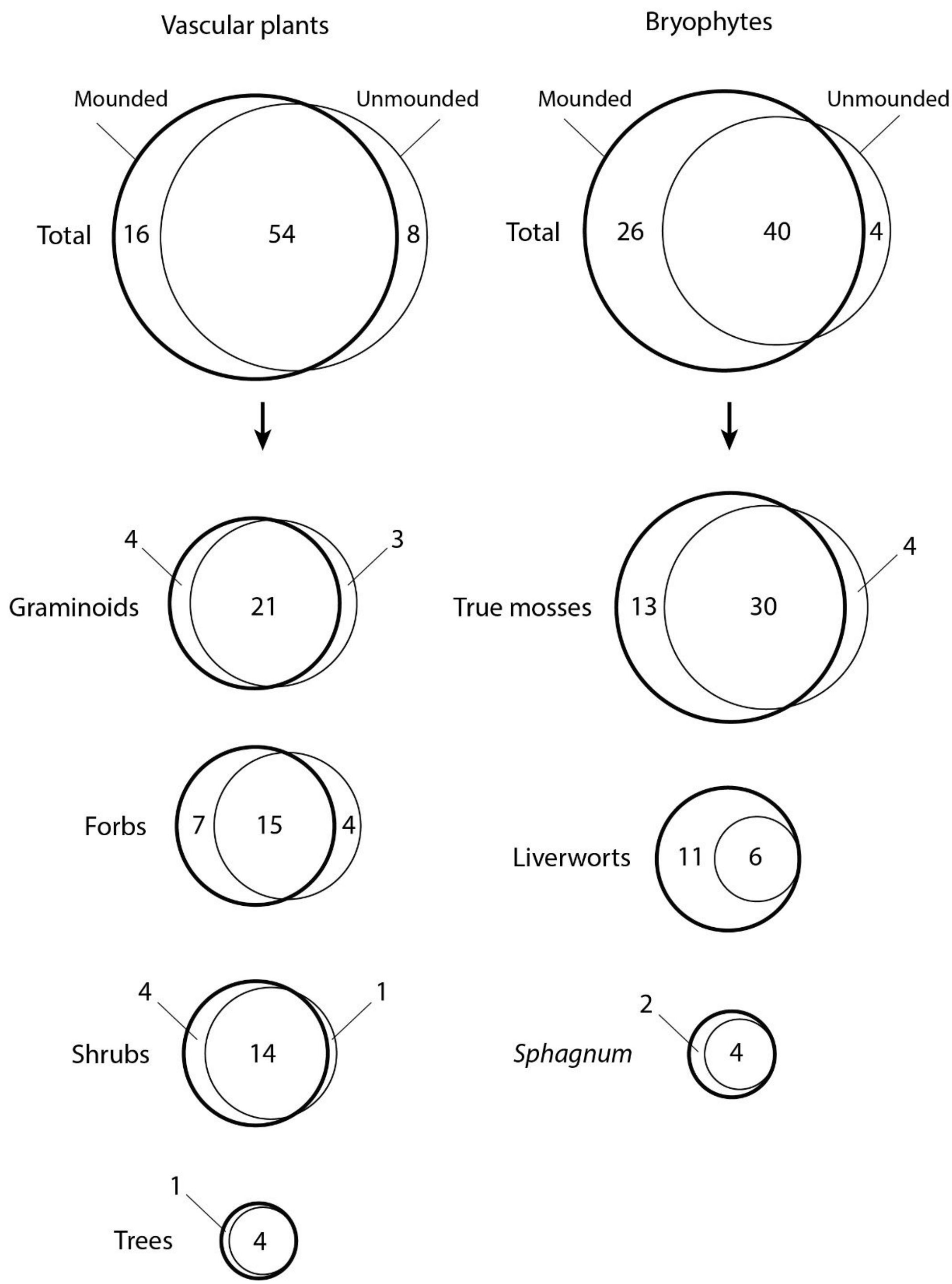
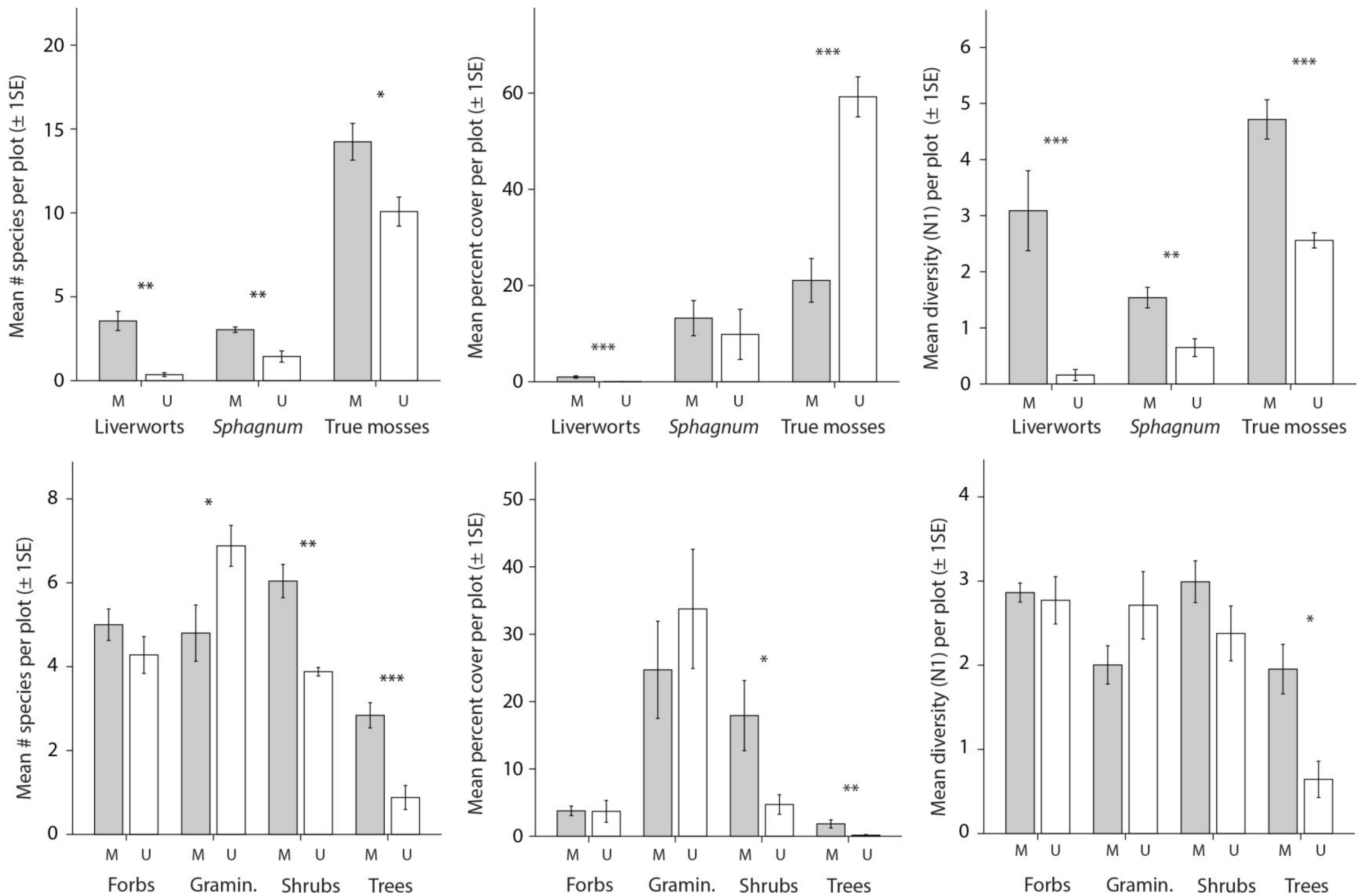


Figure 3.


Vascular plants
Bryophytes
Mounded
Unmounded
Mounded
Unmounded
Total
16
54
8
Total
26
40
4
4
3
Graminoids
21
True mosses
13
30
4
Forbs
7
15
4
Liverworts
11
6
4
1
Shrubs
14
2
Sphagnum
4
1
Trees
4

669 Figure 4.




670

671

Figure 5.

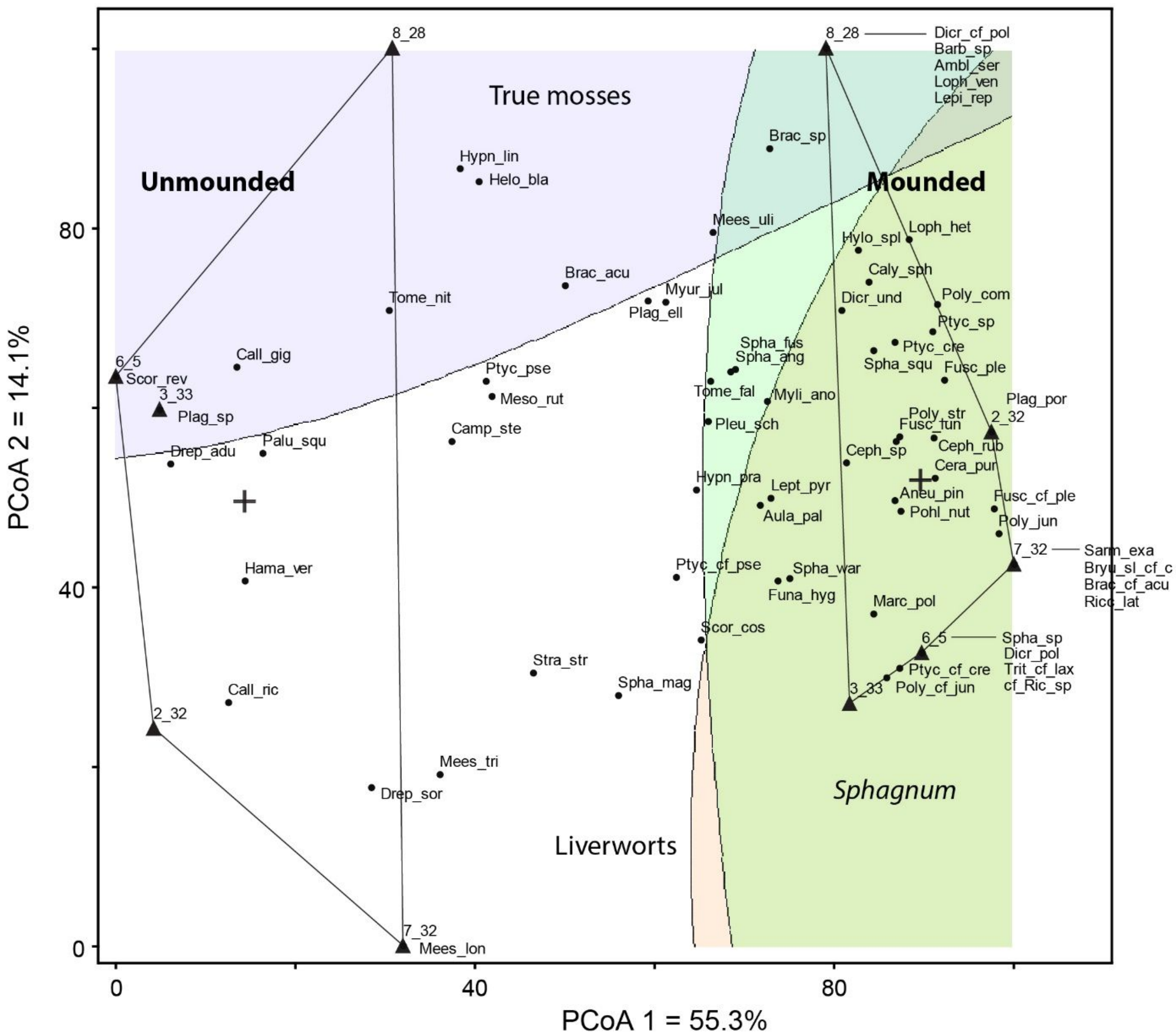
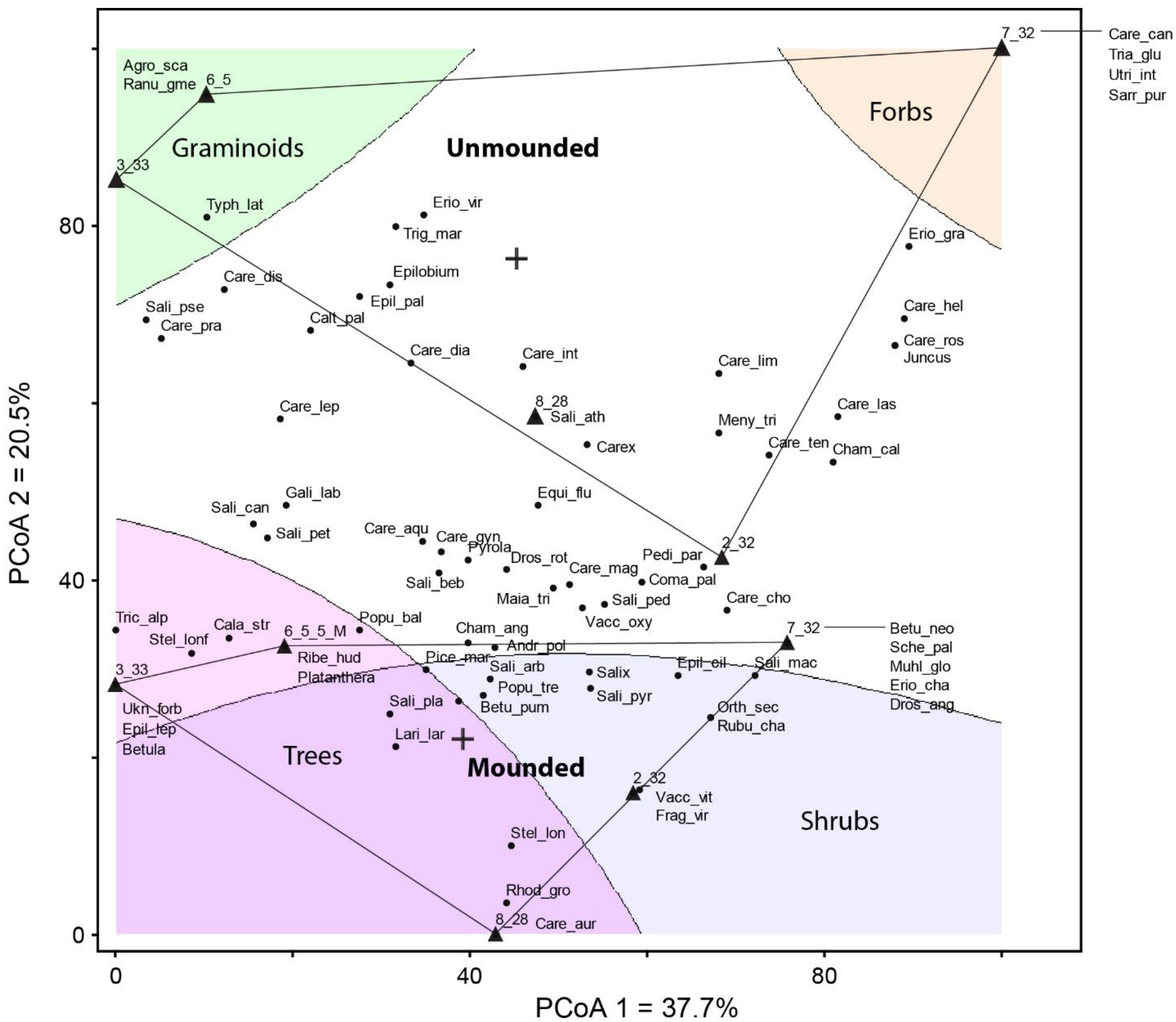

True mosses
Unmounded
Mounded
Liverworts
Sphagnum
PCoA 1 = 55.3%
PCoA 2 = 14.1%
0
40
80
8_28
6_5
3_33
2_32
7_32
Dicr_cf_pol
Barb_sp
Ambl_ser
Loph_ven
Lepi_rep
Brac_sp
Hypn_lin
Helo_bla
Mees_uli
Loph_het
Hylo_spl
Caly_sph
Brac_acu
Tome_nit
Myur_jul
Plag_ell
Dicr_und
Poly_com
Ptyc_sp
Spha_fus
Spha_ang
Ptyc_cre
Spha_squ
Fusc_ple
Call_gig
Scor_rev
Plag_sp
Ptyc_pse
Meso_rut
Tome_fal
Myli_ano
Plag_por
Poly_str
Fusc_fun
Pleu_sch
Ceph_rub
Camp_ste
Drep_adu
Palu_squ
Ceph_sp
Cera_pur
Hypn_pra
Lept_pyr
Aneu_pin
Fusc_cf_ple
Aula_pal
Pohl_nut
Poly_jun
Samn_exa
Bryu_sl_cf_c
Brac_cf_acu
Ricc_lat
Hama_ver
Ptyc_cf_pse
Spha_war
Funa_hyg
Marc_pol
Scor_cos
Spha_sp
Dicr_pol
Trit_cf_lax
cf_Ric_sp
Stra_str
Ptyc_cf_cre
Poly_cf_jun
Call_ric
Spha_mag
Mees_tri
Drep_sor
Mees_lon

Figure 6.


Graminoids
Unmounded
Forbs
Trees
Mounded
Shrubs
7_32
Care_can
Tria_glu
Utri_int
Sarr_pur
Agro_sca
Ranu_gme
6_5
3_33
Typh_lat
Erio_vir
Trig_mar
Erio_gra
Epilobium
Care_dis
Epil_pal
Sali_pse
Care_pra
Calt_pal
Care_hel
Care_ros
Juncus
Care_dia
Care_int
Care_lim
Care_lep
8_28
Sali_ath
Care_las
Meny_tri
Carex
Care_ten
Cham_cal
Gali_lab
Equi_flu
Sali_can
Sali_pet
Care_aqu
Care_gyn
Pyrola
2_32
Pedi_par
Dros_rot
Care_mag
Sali_beb
Coma_pal
Maia_tri
Sali_ped
Care_cho
Vacc_oxy
Tric_alp
Cala_str
Popu_bal
Stel_lonf
6_5_5_M
Cham_ang
7_32
Betu_neo
Sche_pal
Muhl_glo
Erio_cha
Dros_ang
Andr_pol
Ribe_hud
Platanthera
Pice_mar
Sali_arb
Salix
Epil_cil
Sali_mac
3_33
Popu_tre
Sali_pyr
Sali_pla
Betu_pum
Orth_sec
Ukn_forb
Epil_lep
Rubu_cha
Betula
Lari_lar
2_32
Vacc_vit
Frag_vir
Stel_lon
Rhod_gro
8_28
Care_aur
PCoA 2 = 20.5%
PCoA 1 = 37.7%
80
40
0
0
40
80

**Appendix Table A1**. Mean frequency of bryophyte and vascular plant species documented within mounded and unmounded plots for the n = 5 drilling pads. Species are sorted alphabetically by growth form group.

| Species name | Family | Growth form group | Species code | Freq. mounded | Freq. unmounded |
|---|---|---|---|---|---|
| Bryophytes | | | | | |
| *Aneura pinguis* | Aneuraceae | liverwort | Aneu_pin | 0.48 | 0.04 |
| *Calypogeia sphagnicola* | Calypogeiaceae | liverwort | Caly_sph | 0.20 | 0.00 |
| *Fuscocephaloziopsis* cf. *pleniceps* | Cephaloziaceae | liverwort | Fusc_cf_ple | 0.12 | 0.00 |
| *Fuscocephaloziopsis lunulifolia* | Cephaloziaceae | liverwort | Fusc_lun | 0.40 | 0.04 |
| *Fuscocephaloziopsis pleniceps* | Cephaloziaceae | liverwort | Fusc_ple | 0.32 | 0.00 |
| *Cephaloziella rubella* | Cephaloziellaceae | liverwort | Ceph_rub | 0.44 | 0.00 |
| *Cephalozia* s.l. sp. | Cephaloziaceae | liverwort | Ceph_sp | 0.28 | 0.04 |
| cf. *Riccardia* sp. | Aneuraceae | liverwort | cf_Ric_sp | 0.04 | 0.00 |
| *Lepidozia reptans* | Lepidoziaceae | liverwort | Lepi_rep | 0.04 | 0.00 |
| *Tritomaria* cf. *laxa* | Jungermanniaceae | liverwort | Trit_cf_lax | 0.04 | 0.00 |
| *Lophocolea heterophylla* | Lophocoleaceae | liverwort | Loph_het | 0.16 | 0.00 |
| *Mesoptychia rutheana* | Jungermanniaceae | liverwort | Meso_rut | 0.04 | 0.04 |
| *Lophozia ventricosa* | Jungermanniaceae | liverwort | Loph_ven | 0.04 | 0.00 |
| *Marchantia polymorpha* | Marchantiaceae | liverwort | Marc_pol | 0.76 | 0.16 |

| | | | | | |
|---|---|---|---|---|---|
| *Mylia anomala* | Jungermanniaceae | liverwort | Myli_ano | 0.12 | 0.04 |
| *Plagiochila porelloides* | Plagiochilaceae | liverwort | Plag_por | 0.04 | 0.00 |
| *Riccardia latifrons* | Aneuraceae | liverwort | Ricc_lat | 0.04 | 0.00 |
| *Amblystegium serpens* | Amblystegiaceae | true moss | Ambl_ser | 0.04 | 0.00 |
| *Aulacomnium palustre* | Aulacomniaceae | true moss | Aula_pal | 1.00 | 1.00 |
| *Barbula* sp. | Pottiaceae | true moss | Barb_sp | 0.04 | 0.00 |
| *Brachythecium* cf. *acutum* | Brachytheciaceae | true moss | Brac_cf_acu | 0.04 | 0.00 |
| *Brachythecium acutum* | Brachytheciaceae | true moss | Brac_acu | 0.16 | 0.28 |
| *Brachythecium* s.l. sp. | Brachytheciaceae | true moss | Brac_sp | 0.20 | 0.04 |
| *Bryum* s.l. cf. *caespiticium* | Bryaceae | true moss | Bryu_sl_cf_cae | 0.08 | 0.00 |
| *Calliergon giganteum* | Calliergonaceae | true moss | Call_gig | 0.12 | 0.36 |
| *Calliergon richardsonii* | Calliergonaceae | true moss | Call_ric | 0.04 | 0.16 |
| *Campylium stellatum* | Amblystegiaceae | true moss | Camp_ste | 0.48 | 0.72 |
| *Ceratodon purpureus* | Ditrichaceae | true moss | Cera_pur | 1.00 | 0.16 |
| *Dicranum* cf. *polysetum* | Dicranaceae | true moss | Dicr_cf_pol | 0.04 | 0.00 |
| *Dicranum polysetum* | Dicranaceae | true moss | Dicr_pol | 0.04 | 0.00 |
| *Dicranum undulatum* | Dicranaceae | true moss | Dicr_und | 0.68 | 0.08 |
| *Drepanocladus aduncus* | Amblystegiaceae | true moss | Drep_adu | 0.20 | 0.52 |
| *Drepanocladus sordidus* | Amblystegiaceae | true moss | Drep_sor | 0.00 | 0.12 |
| *Funaria hygrometrica* | Funariaceae | true moss | Funa_hyg | 0.68 | 0.16 |
| *Hamatocaulis vernicosus* | Calliergonaceae | true moss | Hama_ver | 0.40 | 0.96 |
| *Elodium blandowii* | Helodiaceae | true moss | Helo_bla | 0.36 | 0.28 |

| | | | | | |
|---|---|---|---|---|---|
| *Hylocomium splendens* | Hylocomiaceae | true moss | Hylo_spl | 0.12 | 0.00 |
| *Hypnum lindbergii* | Hypnaceae | true moss | Hypn_lin | 0.04 | 0.08 |
| *Hypnum pratense* | Hypnaceae | true moss | Hypn_pra | 0.36 | 0.16 |
| *Leptobryum pyriforme* | Meesiaceae | true moss | Lept_pyr | 0.80 | 0.24 |
| *Scorpidium revolvens* | Calliergonaceae | true moss | Scor_rev | 0.00 | 0.08 |
| *Meesia longiseta* | Meesiaceae | true moss | Mees_lon | 0.00 | 0.04 |
| *Meesia triquetra* | Meesiaceae | true moss | Mees_tri | 0.36 | 0.64 |
| *Meesia uliginosa* | Meesiaceae | true moss | Mees_uli | 0.20 | 0.12 |
| *Myurella julacea* | Pterigynandraceae | true moss | Myur_jul | 0.12 | 0.04 |
| *Paludella squarrosa* | Meesiaceae | true moss | Palu_squ | 0.08 | 0.20 |
| *Plagiomnium ellipticum* | Mniaceae | true moss | Plag_ell | 0.48 | 0.24 |
| *Plagiomnium* sp. | Mniaceae | true moss | Plag_sp | 0.00 | 0.04 |
| *Pleurozium schreberi* | Hylocomiaceae | true moss | Pleu_sch | 0.40 | 0.16 |
| *Pohlia nutans* | Mielichhoferiaceae | true moss | Pohl_nut | 0.88 | 0.16 |
| *Polytrichum* cf. *juniperinum*[1] | Polytrichaceae | true moss (for analysis) | Poly_cf_jun | 0.08 | 0.00 |
| *Polytrichum commune*[1] | Polytrichaceae | true moss (for analysis) | Poly_com | 0.12 | 0.00 |
| *Polytrichum juniperinum*[1] | Polytrichaceae | true moss (for analysis) | Poly_jun | 0.44 | 0.00 |
| *Polytrichum strictum*[1] | Polytrichaceae | true moss (for analysis) | Poly_str | 0.96 | 0.40 |
| *Ptychostomum* cf. *creberrimum* | Bryaceae | true moss | Ptyc_cf_cre | 0.16 | 0.00 |
| *Ptychostomum* cf. *pseudotriquetrum* | Bryaceae | true moss | Ptyc_cf_pse | 0.08 | 0.04 |
| *Ptychostomum creberrimum* | Bryaceae | true moss | Ptyc_cre | 0.60 | 0.08 |
| *Ptychostomum pseudotriquetrum* | Bryaceae | true moss | Ptyc_pse | 0.52 | 0.88 |

| | | | | | |
|---|---|---|---|---|---|
| *Ptychostomum* sp. | Bryaceae | true moss | Ptyc_sp | 0.24 | 0.00 |
| *Sarmentypnum exannulatum* | Calliergonaceae | true moss | Sarm_exa | 0.04 | 0.00 |
| *Scorpidium cossonii* | Calliergonaceae | true moss | Scor_cos | 0.04 | 0.08 |
| *Straminergon stramineum* | Calliergonaceae | true moss | Stra_str | 0.20 | 0.36 |
| *Tomentypnum falcifolium* | Amblystegiaceae | true moss | Tome_fal | 0.36 | 0.24 |
| *Tomentypnum nitens* | Amblystegiaceae | true moss | Tome_nit | 0.96 | 0.96 |
| *Sphagnum angustifolium* | Sphagnaceae | *Sphagnum* | Spha_ang | 0.88 | 0.36 |
| *Sphagnum fuscum* | Sphagnaceae | *Sphagnum* | Spha_fus | 0.52 | 0.12 |
| *Sphagnum magellanicum* | Sphagnaceae | *Sphagnum* | Spha_mag | 0.60 | 0.24 |
| *Sphagnum* sp. | Sphagnaceae | *Sphagnum* | Spha_sp | 0.04 | 0.00 |
| *Sphagnum squarrosum* | Sphagnaceae | *Sphagnum* | Spha_squ | 0.08 | 0.00 |
| *Sphagnum warnstorfii* | Sphagnaceae | *Sphagnum* | Spha_war | 0.92 | 0.72 |
| Vascular plants | | | | | |
| *Caltha palustris* | Ranunculaceae | forb | Calt_pal | 0.12 | 0.20 |
| *Chamerion angustifolium* | Onagraceae | forb | Cham_ang | 0.40 | 0.04 |
| *Comarum palustre* | Rosaceae | forb | Coma_pal | 0.48 | 0.56 |
| *Drosera anglica* | Droseraceae | forb | Dros_ang | 0.08 | 0.00 |
| *Drosera rotundifolia* | Droseraceae | forb | Dros_rot | 0.68 | 0.28 |
| *Epilobium ciliatum* | Onagraceae | forb | Epil_cil | 0.04 | 0.04 |
| *Epilobium leptophyllum* | Onagraceae | forb | Epil_lep | 0.04 | 0.00 |
| *Epilobium palustre* | Onagraceae | forb | Epil_pal | 0.04 | 0.32 |

| | | | | | |
|---|---|---|---|---|---|
| *Epilobium* sp. | Onagraceae | forb | Epilobium | 0.08 | 0.24 |
| *Equisetum fluviatile* | Equisetaceae | forb | Equi_flu | 0.52 | 0.52 |
| *Fragaria virginiana* | Rosaceae | forb | Frag_vir | 0.04 | 0.00 |
| *Galium labradoricum* | Rubiaceae | forb | Gali_lab | 0.16 | 0.08 |
| *Maianthemum trifolium* | Asparagaceae | forb | Maia_tri | 0.80 | 0.64 |
| *Menyanthes trifoliata* | Menyantheaceae | forb | Meny_tri | 0.64 | 0.68 |
| *Orthilia secunda* | Ericaceae | forb | Orth_sec | 0.08 | 0.00 |
| *Pedicularis parviflora* | Orobanchaceae | forb | Pedi_par | 0.08 | 0.04 |
| *Platanthera* sp. | Orchidaceae | forb | Platanthera | 0.04 | 0.00 |
| *Pyrola* sp. | Ericaceae | forb | Pyrola | 0.08 | 0.08 |
| *Ranunculus gmelinii* | Ranunculaceae | forb | Ranu_gme | 0.00 | 0.04 |
| *Rubus chamaemorus* | Rosaceae | forb | Rubu_cha | 0.08 | 0.00 |
| *Sarracenia purpurea* | Sarraceniaceae | forb | Sarr_pur | 0.00 | 0.12 |
| *Stellaria* cf. *longifolia* | Caryophyllaceae | forb | Stel_cf._lof | 0.00 | 0.00 |
| *Stellaria longipes* | Caryophyllaceae | forb | Stel_lon | 0.16 | 0.16 |
| *Stellaria longifolia* | Caryophyllaceae | forb | Stel_lonf | 0.32 | 0.12 |
| *Triantha glutinosa* | Tofieldiaceae | forb | Tria_glu | 0.00 | 0.04 |
| *Utricularia intermedia* | Lentibulariaceae | forb | Utri_int | 0.00 | 0.08 |
| *Calamagrostis stricta* ssp. *stricta* | Poaceae | graminoid | Cala_str | 0.16 | 0.04 |
| *Carex aquatilis* | Cyperaceae | graminoid | Care_aqu | 0.96 | 1.00 |
| *Carex aurea* | Cyperaceae | graminoid | Care_aur | 0.04 | 0.00 |
| *Carex canescens* | Cyperaceae | graminoid | Care_can | 0.00 | 0.04 |

| | | | | | |
|---|---|---|---|---|---|
| *Carex chordorrhiza* | Cyperaceae | graminoid | Care_cho | 0.32 | 0.28 |
| *Carex diandra* | Cyperaceae | graminoid | Care_dia | 0.68 | 0.88 |
| *Carex disperma* | Cyperaceae | graminoid | Care_dis | 0.04 | 0.16 |
| *Carex gynocrates* | Cyperaceae | graminoid | Care_gyn | 0.12 | 0.16 |
| *Carex heleonastes* | Cyperaceae | graminoid | Care_hel | 0.16 | 0.12 |
| *Carex interior* | Cyperaceae | graminoid | Care_int | 0.64 | 0.92 |
| *Carex lasiocarpa* | Cyperaceae | graminoid | Care_las | 0.04 | 0.08 |
| *Carex leptalea* | Cyperaceae | graminoid | Care_lep | 0.12 | 0.24 |
| *Carex limosa* | Cyperaceae | graminoid | Care_lim | 0.16 | 0.44 |
| *Carex magellanica* | Cyperaceae | graminoid | Care_mag | 0.08 | 0.28 |
| *Carex prairea* | Cyperaceae | graminoid | Care_pra | 0.16 | 0.32 |
| *Carex rostrata* | Cyperaceae | graminoid | Care_ros | 0.04 | 0.04 |
| *Carex tenuiflora* | Cyperaceae | graminoid | Care_ten | 0.36 | 0.40 |
| *Carex* sp. | Cyperaceae | graminoid | Carex | 0.08 | 0.12 |
| *Agrostis scabra* | Poaceae | graminoid | Agro_sca | 0.00 | 0.04 |
| *Eriophorum chamissonis* | Cyperaceae | graminoid | Erio_cha | 0.04 | 0.00 |
| *Eriophorum gracile* | Cyperaceae | graminoid | Erio_gra | 0.12 | 0.36 |
| *Eriophorum viridi-carinatum* | Cyperaceae | graminoid | Erio_vir | 0.12 | 0.24 |
| *Juncus* sp. | Juncaceae | graminoid | Juncus | 0.04 | 0.04 |
| *Muhlenbergia glomerata* | Poaceae | graminoid | Muhl_glo | 0.04 | 0.00 |
| *Scheuchzeria palustris* | Scheuchzeriaceae | graminoid | Sche_pal | 0.04 | 0.00 |
| *Trichophorum alpinum* | Cyperaceae | graminoid | Tric_alp | 0.08 | 0.04 |

| | | | | | |
|---|---|---|---|---|---|
| *Triglochin maritima* | Juncaginaceae | graminoid | Trig_mar | 0.00 | 0.32 |
| *Typha latifolia* | Typhaceae | graminoid | Typh_lat | 0.16 | 0.32 |
| *Andromeda polifolia* | Ericaceae | shrub | Andr_pol | 0.60 | 0.44 |
| *Betula pumila* | Betulaceae | shrub | Betu_pum | 0.84 | 0.60 |
| *Chamaedaphne calyculata* | Ericaceae | shrub | Cham_cal | 0.16 | 0.08 |
| *Rhododendron groenlandicum* | Ericaceae | shrub | Rhod_gro | 0.20 | 0.08 |
| *Ribes hudsonianum* | Grossulariaceae | shrub | Ribe_hud | 0.04 | 0.00 |
| *Salix arbusculoides* | Salicaceae | shrub | Sali_arb | 0.08 | 0.04 |
| *Salix athabascensis* | Salicaceae | shrub | Sali_ath | 0.00 | 0.04 |
| *Salix bebbiana* | Salicaceae | shrub | Sali_beb | 0.88 | 0.56 |
| *Salix candida* | Salicaceae | shrub | Sali_can | 0.16 | 0.08 |
| *Salix maccalliana* | Salicaceae | shrub | Sali_mac | 0.12 | 0.00 |
| *Salix pedicellaris* | Salicaceae | shrub | Sali_ped | 0.72 | 0.80 |
| *Salix petiolaris* | Salicaceae | shrub | Sali_pet | 0.08 | 0.08 |
| *Salix planifolia* | Salicaceae | shrub | Sali_pla | 0.24 | 0.12 |
| *Salix pseudomyrsinites* | Salicaceae | shrub | Sali_pse | 0.04 | 0.08 |
| *Salix pyrifolia* | Salicaceae | shrub | Sali_pyr | 0.48 | 0.28 |
| *Salix* sp. | Salicaceae | shrub | Salix | 0.64 | 0.24 |
| *Vaccinium oxycoccos* | Ericaceae | shrub | Vacc_oxy | 0.64 | 0.36 |
| *Vaccinium vitis-idaea* | Ericaceae | shrub | Vacc_vit | 0.12 | 0.00 |
| *Betula neoalaskana* | Betulaceae | tree | Betu_neo | 0.04 | 0.00 |
| *Larix laricina* | Pinaceae | tree | Lari_lar | 0.64 | 0.20 |

| | | | | | |
|---|---|---|---|---|---|
| *Picea mariana* | Pinaceae | tree | Pice_mar | 0.48 | 0.12 |
| *Populus balsamifera* | Salicaceae | tree | Popu_bal | 0.72 | 0.28 |
| *Populus tremuloides* | Salicaceae | tree | Popu_tre | 0.92 | 0.28 |
| *Betula* sp. | Betulaceae | tree | Betula | 0.04 | 0.00 |

Notes:

[1]Mosses in the genus *Polytrichum* belong to Class Polytrichopsida and are not true mosses, but have been included in that category to facilitate data interpretation.

**Appendix Table A2**. Results of PERMANOVA for differences in species composition between mounded and unmounded habitat for (A) bryophytes and (B) vascular plants.

(A) Bryophytes

| Source | df | SS | MS | F | p |
|---|---|---|---|---|---|
| Habitat (mounded versus unmounded) | 1 | 0.921 | 0.921 | 8.821 | 0.008 |
| Residual | 8 | 0.835 | 0.104 | | |
| Total | 9 | 1.757 | | | |

(B) Vascular plants

| Source | df | SS | MS | F | p |
|---|---|---|---|---|---|
| Habitat (mounded versus unmounded) | 1 | 0.258 | 0.258 | 1.735 | 0.097 |
| Residual | 8 | 1.190 | 0.149 | | |
| Total | 9 | 1.448 | | | |

**Appendix Table A3**. Results of Indicator Species Analysis for bryophytes and vascular plants, between mounded and unmounded habitat for the n = 5 drilling pads. Species are listed in order of decreasing indicator value (IV). Only significant indicator species are shown.

| Species | Growth form group | Mounded versus unmounded habitat | IV | p |
|---|---|---|---|---|
| Bryophytes | | | | |
| *Hamatocaulis vernicosus* | true moss | unmounded | 99.9 | 0.008 |
| *Ptychostomum creberrimum* | true moss | mounded | 99.1 | 0.008 |
| *Ceratodon purpureus* | true moss | mounded | 99.0 | 0.008 |
| *Marchantia polymorpha* | liverwort | mounded | 97.5 | 0.024 |
| *Polytrichum strictum*[1] | true moss (for analysis) | mounded | 96.7 | 0.008 |
| *Dicranum undulatum* | true moss | mounded | 92.9 | 0.015 |
| *Pohlia nutans* | true moss | mounded | 92.6 | 0.008 |
| *Aneura pinguis* | liverwort | mounded | 92.3 | 0.023 |
| *Funaria hygrometrica* | true moss | mounded | 86.6 | 0.039 |
| *Cephalozia* s.l. sp. | liverwort | mounded | 86.3 | 0.022 |
| *Aulacomnium palustre* | true moss | mounded | 81.7 | 0.008 |
| *Fuscocephaloziopsis pleniceps* | liverwort | mounded | 80.0 | 0.047 |
| *Polytrichum juniperinum*[1] | true moss (for analysis) | mounded | 80.0 | 0.047 |
| *Cephaloziella rubella* | liverwort | mounded | 80.0 | 0.049 |
| *Fuscocephaloziopsis lunulifolia* | liverwort | mounded | 76.0 | 0.048 |
| Vascular plants | | | | |
| *Populus tremuloides* | tree | mounded | 96.5 | 0.008 |
| *Betula pumila* | shrub | mounded | 95.2 | 0.008 |

| | | | | |
|---|---|---|---|---|
| *Salix* spp. | shrub | mounded | 94.4 | 0.024 |
| *Populus balsamifera* | tree | mounded | 89.4 | 0.032 |
| *Picea mariana* | tree | mounded | 80.0 | 0.038 |
| *Triglochin maritima* | forb | unmounded | 80.0 | 0.049 |

Notes:

[1]*Polytrichum juniperinum* and *P. strictum* belong to Class Polytrichopsida and are not true mosses, but have been included in that category to facilitate data interpretation.